

DI WU

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EDUCATION

University of California, Los Angeles

Los Angeles, CA, USA

Ph.D. Candidate in Computer Science (Advisor: Prof. Kai-Wei Chang)

09/2022 - present

- Overall GPA: **4.000/4.000**
- Research Interests: long-horizon agents, retrieval-augmented language models, text generation evaluation.

University of California, Los Angeles

Los Angeles, CA, USA

B.S. in Computer Science, Summa Cum Laude

08/2018 - 06/2022

- Overall GPA: **3.973/4.000**

RESEARCH EXPERIENCE

UCLA NLP Lab

Los Angeles, CA, USA

Student Researcher, Supervisor: Prof. Kai-Wei Chang

02/2021 - present

- Research on neural keyphrase generation methods and their evaluation.
- Research on building robust retrieval-augmented generation and long-term memory methods.
- Collaborate with Taboola on a number of real-world systems including concept extraction, continual learning, knowledge agents, and automatic content moderation.
- Collaborate with Prof. Tian Yuan's team on building robust coding agents for Amazon Nova Challenge.

The Ozcan Research Group, UCLA

Los Angeles, CA, USA

Student Research Assistant, Supervisor: Prof. Aydogan Ozcan

09/2019 - 06/2022

- Researched on virtual staining of skin tissues with an emphasis on Basal Cell Carcinoma. Designed methods to improve both single-image quality and temporal coherence of the predictions.
- Explored dataset engineering and data augmentation methods to mitigate class imbalance.

PUBLICATIONS

Wu, D., Ji, Z., Kawatkar, A., Kwan, B., Gu, J. C., Peng, N., Chang, K. W., 2026. LongMemEval-V2: Evaluating Long-Term Agent Memory Toward Experienced Colleagues. *preprint, ICML 2026 SCALE Workshop (Oral)*.

Wu, D., Sachan, D. S., Yih, W. T., Chen, M., 2026. Procedural Knowledge at Scale Improves Reasoning. *COLM 2026*.

Wu, D.*, Wan, Y.*, Chang, K. W., 2025. VisRet: Visualization Improves Knowledge-Intensive Text-to-Image Retrieval. *ACL 2026 (Oral)*.

Wu, D.*, Gu, J. C.*, Chang, K. W., Peng, N., 2025. Self-Routing RAG: Binding Selective Retrieval with Knowledge Verbalization. *preprint*.

Gu, J. C., Zhang, J., **Wu, D.**, Li, Y., Chang, K. W., Peng, N., 2025. BRIEF-Pro: Universal Context Compression with Short-to-Long Synthesis for Fast and Accurate Multi-Hop Reasoning. *Findings of the ACL: ACL 2026*.

Wu, D., Liu, S., Ji, Z., Chang, Y.-L., Liu, Z.-Y., Pleffer, A., Chang, K. W., 2025. Open-Domain Safety Policy Construction. *Findings of the ACL: EACL 2026*.

Wang, Y., Yin, D., Cui, Y., Li, Z., Zheng, R., Lin, Z., **Wu, D.**, Wu, X., Ye, C., Zhou, Y., Chang, K. W., 2025. LLMs as Scalable, General-Purpose Simulators for Evolving Digital Agent Training. *NeurIPS 2025 LAW Workshop*.

Li, Y., Gu, J. C., **Wu, D.**, Chang, K. W., Peng, N., 2024. BRIEF: Bridging Retrieval and Inference via Multi-hop Reasoning and Compression. *Findings of the ACL: NAACL 2025*.

- Wu, D.**, Wang, H., Yu, W., Zhang, Y., Chang, K. W., Yu, D., 2024. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. *ICLR 2025*.
- Wu, D.**, Gu, J. C., Yin, F., Peng, N., and Chang, K. W., 2024. Synchronous Faithfulness Monitoring for Trustworthy Retrieval-Augmented Generation. *EMNLP 2024*.
- Wan, Y., **Wu, D.**, Wang, H., and Chang, K. W., 2024. The Factuality Tax of Diversity-Intervened Text-to-Image Generation: Benchmark and Fact-Augmented Intervention. *EMNLP 2024*.
- Wu, D.**, Shen, X., and Chang, K. W., 2024. MetaKP: On-Demand Keyphrase Generation. *Findings of the ACL: EMNLP 2024*.
- Wu, D.**, Yin, D., and Chang, K. W., 2024. KPEval: Towards Fine-grained Semantic-based Keyphrase Evaluation. *Findings of the ACL: ACL 2024*.
- Wu, D.**, Ahmad, W. U., Zhang, D., Ramanathan, M. K., and Ma, X., 2023. Repoformer: Selective Retrieval for Repository-level Code Completion. *ICML 2024 (Oral)*.
- Wu, D.**, Ahmad, W. U., and Chang, K. W., 2023. On Leveraging Encoder-only Pre-trained Language Models for Effective Keyphrase Generation. *LREC-COLING 2024*.
- Li, Y., Pillar, N., Li, J., Liu, T., **Wu, D.**, Sun, S., Ma, G., de Haan, K., Huang, L., Zhang, Y. and Hamidi, S., 2024. Virtual histological staining of unlabeled autopsy tissue. *Nature Communications*, 15(1), p.1684.
- Wu, D.**, Ahmad, W. U., and Chang, K. W., 2023. Rethinking Model Selection and Decoding for Keyphrase Generation with Pre-trained Sequence-to-Sequence Models. *EMNLP 2023*.
- Kung, P., Yin F., **Wu, D.**, Chang, K. W., and Peng N., 2023. Active Instruction Tuning: Improving Cross-Task Generalization by Training on Prompt Sensitive Tasks. *EMNLP 2023*.
- Wu, D.**, Ahmad, W. U., and Chang, K. W., 2023. Pre-trained Language Models for Keyphrase Generation: A Thorough Empirical Study. *Preprint*.
- Wu, D.**, Ahmad, W. U., Dev, S., and Chang, K. W., 2023. Representation Learning for Resource-Constrained Keyphrase Generation. *Findings of the ACL: EMNLP 2022*.
- Li, J., Garfinkel, J., Zhang, X., **Wu, D.**, Zhang, Y., de Haan, K., Wang, H., Liu, T., Bai, B., Rivenson, Y., Rubinstein, G., Scumpia, P., and Ozcan, A., 2023. Biopsy-free in vivo virtual histology of skin using deep learning. *Light: Science & Applications*, 10(1), 1-22.

INTERNSHIP EXPERIENCE

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|---|--------------------|
| Google Cloud AI Research | Sunnyvale, CA, USA |
| <i>Student Researcher; Mentors: Rujun Han, Yanfei Chen, Chen-Yu Lee</i> | 06/2026 - 09/2026 |
| <ul style="list-style-type: none"> • Researching on continually evolving multi-agent systems. | |
| FAIR at Meta Superintelligence Labs | New York, NY, USA |
| <i>Research Scientist Intern; Mentors: Mingda Chen, Devendra Sachan, Scott Yih</i> | 06/2025 - 10/2025 |
| <ul style="list-style-type: none"> • Researched on memory representations for improving frontier reasoning tasks. • Built the first large-scale procedural memory datastore that significantly improves various reasoning models. | |
| Tencent AI Lab | Bellevue, WA, USA |
| <i>Research Intern; Mentors: Hongwei Wang, Wenhao Yu</i> | 06/2024 - 09/2024 |
| <ul style="list-style-type: none"> • Researched on long-term memory for chat assistants and built a large-scale high-quality evaluation benchmark. • Designed indexing strategies for improving long-term memory retrieval performance. | |
| AWS AI Labs | New York, NY, USA |
| <i>Applied Scientist Intern; Mentors: Wasi Ahmad, Dejiao Zhang</i> | 06/2023 - 09/2023 |
| <ul style="list-style-type: none"> • Researched on improving retrieval-augmented code language models for repository-level code completion. | |

- Formulated the task of selective retrieval-augmented infilling. Designed approaches from the perspective of in-repository code retrievers and the code generator models.
- The designed model achieves no performance loss with only 10% of retrieval budget.

Microsoft Research Asia

Research Intern; Mentor: Ning Shang

Beijing, China
04/2021 - 09/2021

- Researched on model compression and AutoML. Contributed to the open source project [NNI](#).

NewsBreak

Natural Language Processing Intern

Beijing, China
07/2020 - 10/2020

- Worked on a hierarchical multi-label classification problem with 268 categories. Improved the f1-score by 49% with multiple statistical and deep learning methods.
- Improved the performance of model pre-training, fine-tuning, and online serving pipelines.

TEACHING

Teaching Assistant

- UCLA CS 33, Introduction to Computer Organization, Fall 2023, with Prof. Glenn Reinman.
- UCLA CS 33, Introduction to Computer Organization, Spring 2024, with Prof. Glenn Reinman.
- UCLA CS 33, Introduction to Computer Organization, Fall 2024, with Prof. Tony Nowatzki.

SERVICES AND AWARDS

- Reviewer: ACL 2023, EMNLP 2023, AAAI, ICLR, ICML, NeurIPS, JAIR, ACL Rolling Review.
- Outstanding Reviewer, EMNLP 2024.
- Amazon Trainium Fellowship, Fall 2025.
- UCLA HSSEAS Dean's List, 8 times, 2018-2022.